

## 2.5 Row Space, Column Space, Null Space and Rank

We can now reduce any matrix and read off the solutions of any linear system. This section asks a coarser question: how much information does a matrix carry? Three subspaces come attached to every matrix  $A$ : the vectors  $A\mathbf{x}$  it can produce, the combinations of its rows, and the vectors it sends to zero. Their dimensions are governed by a single number, the rank, which is the number of pivots. With it we settle when a system is solvable and how many solutions it has, and we turn the questions of Chapter 1 about vectors in  $F^m$  (independent? spanning? a basis of a sum or an intersection?) into a routine computation, which is the “systematic method” that Chapter 1 promised.

Throughout,  $F$  is a field and  $A \in M_{m \times n}(F)$  has columns  $\mathbf{a}_1, \dots, \mathbf{a}_n \in F^m$ .

### 2.5.1 Three subspaces of a matrix

Here is the matrix we will follow through the section:

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

Three questions about it have come up in this chapter, and each answer is a set. For which  $\mathbf{b} \in \mathbb{R}^4$  can we solve  $A\mathbf{x} = \mathbf{b}$ ? By Theorem 2.1.1, exactly for  $\mathbf{b}$  in the span of the columns. Which  $\mathbf{x} \in \mathbb{R}^5$  solve  $A\mathbf{x} = \mathbf{0}$ ? By Theorem 2.1.2, a subspace of  $\mathbb{R}^5$ . And which new equations can we derive from the four by adding multiples of them? Their left-hand sides are exactly the combinations of the rows. These three sets recur so often that they get names.

*The column space is everything  $A$  can produce, the null space is everything  $A$  kills, and the row space is every left-hand side we can build from the equations of  $A\mathbf{x} = \mathbf{b}$ .*

#### Definition 2.5.1 — Column space

The **column space** of  $A \in M_{m \times n}(F)$  is the span of its columns,

$$\text{col}(A) := \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \{A\mathbf{x} : \mathbf{x} \in F^n\} \subseteq F^m.$$

#### Definition 2.5.2 — Row space

The **row space** of  $A \in M_{m \times n}(F)$  is the span of its rows,

$$\text{row}(A) := \text{span}(\text{row } 1, \dots, \text{row } m) \subseteq M_{1 \times n}(F).$$

#### Definition 2.5.3 — Null space

The **null space** of  $A \in M_{m \times n}(F)$  is the solution set of the homogeneous system,

$$\text{nul}(A) := \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\} \subseteq F^n.$$

In words:  $\text{col}(A)$  lives among the **outputs**, in  $F^m$  (one entry per row);  $\text{nul}(A)$  lives among the **inputs**, in  $F^n$  (one entry per column); and  $\text{row}(A)$  consists of row vectors of length  $n$ . The two descriptions of  $\text{col}(A)$  agree by Theorem 0.9.3:  $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ , and as  $\mathbf{x}$  runs over  $F^n$  these are all the combinations of the columns.

**Well-definedness.** All three are subspaces.  $\text{col}(A)$  and  $\text{row}(A)$  are spans, hence subspaces by Theorem 1.3.1, of  $F^m$  and of  $M_{1 \times n}(F)$  respectively.  $\text{nul}(A)$  is a subspace of  $F^n$  by Theorem 2.1.2. The row vectors  $M_{1 \times n}(F)$  form an  $n$ -dimensional space, with the standard basis  $\mathbf{e}_1^T, \dots, \mathbf{e}_n^T$  (Example (Dimensions of the Running Examples)). Transposing turns row vectors into columns and respects sums and scalar multiples (Theorem 0.9.5), so it sends a combination of rows to the same combination of their transposes, and only the zero row transposes to  $\mathbf{0}$ . The transposes of the rows of  $A$  are the columns of  $A^T$ . Hence transposing sends  $\text{row}(A)$  onto  $\text{col}(A^T)$ , a list of rows is independent exactly when the list of their transposes is, and transposing a basis of  $\text{row}(A)$  gives a basis of  $\text{col}(A^T)$ . Therefore

$$\dim \text{row}(A) = \dim \text{col}(A^T). \quad (*)$$

### Examples.

- **The identity.**  $\text{col}(I_n) = F^n$ , since the columns  $\mathbf{e}_1, \dots, \mathbf{e}_n$  span;  $\text{row}(I_n) = M_{1 \times n}(F)$ ; and  $\text{nul}(I_n) = \{\mathbf{0}\}$ , since  $I_n\mathbf{x} = \mathbf{x}$ .
- **The zero matrix** is the degenerate case. For  $0 = 0_{m \times n}$ ,  $\text{col}(0) = \text{span}(\mathbf{0}, \dots, \mathbf{0}) = \{\mathbf{0}\}$ ,  $\text{row}(0) = \{\mathbf{0}\}$ , and  $\text{nul}(0) = F^n$ , since every  $\mathbf{x}$  is killed. It carries no information, and its column space is as small as possible while its null space is as large as possible. We will see that this trade-off is exact.
- **A single row.** For  $A = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \in M_{1 \times 3}(\mathbb{R})$ , the columns are the scalars  $1, 1, -1 \in \mathbb{R}^1$ , so  $\text{col}(A) = \mathbb{R}^1$ ,  $\text{row}(A) = \text{span}(\begin{pmatrix} 1 & 1 & -1 \end{pmatrix})$  is a line of row vectors, and  $\text{nul}(A)$  is the plane  $x + y - z = 0$  in  $\mathbb{R}^3$ .

**Non-example by minimal change.** Replace  $\mathbf{0}$  by a non-zero right-hand side: the solution set  $\{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{b}\}$  with  $\mathbf{b} \neq \mathbf{0}$ . It is described by the same equations, and it is still of the form  $\mathbf{p} + \text{nul}(A)$  when non-empty (Theorem 2.1.3). But it does not contain  $\mathbf{0}$ , since  $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ , so it fails the first condition of Theorem 1.2.1.

*Warning.* The three spaces live in **different places**. For  $A \in M_{4 \times 5}(\mathbb{R})$  above,  $\text{col}(A) \subseteq \mathbb{R}^4$  but  $\text{nul}(A) \subseteq \mathbb{R}^5$ , so a statement such as “ $\text{nul}(A) \subseteq \text{col}(A)$ ” does not even make sense. Keep track by size:  $A\mathbf{x}$  needs  $\mathbf{x}$  with  $n$  entries and returns a vector with  $m$  entries.

## 2.5.2 Rank and nullity

The column space measures how much  $A$  can reach, and the null space how much it collapses. We measure each by its dimension. Both are subspaces of finite-dimensional spaces, so both have a dimension by Theorem 1.6.7.

*The rank counts the independent directions that  $A$  can produce.*

**Definition 2.5.4 — Rank of a matrix**

The **rank** of  $A \in M_{m \times n}(F)$  is  $\text{rank } A := \dim \text{col}(A)$ .

**Definition 2.5.5 — Nullity of a matrix**

The **nullity** of  $A \in M_{m \times n}(F)$  is  $\text{nullity } A := \dim \text{nul}(A)$ .

From the examples above:  $\text{rank } I_n = n$  and  $\text{nullity } I_n = 0$ ;  $\text{rank } 0_{m \times n} = 0$  and  $\text{nullity } 0_{m \times n} = n$ ; and  $\text{rank} \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} = 1$ , with  $\text{nullity } 2$ . In each case rank plus nullity equals the number of columns. Two simple bounds hold at once:  $\text{rank } A \leq m$ , because  $\text{col}(A) \subseteq F^m$ , and  $\text{rank } A \leq n$ , because  $\text{col}(A)$  is spanned by  $n$  vectors (Theorem 1.6.3). Why define rank through columns and not rows? We could have, and one of the main theorems of this section says the answer would be the same.

**2.5.3 What row operations preserve**

To compute these spaces we will row reduce. So we first ask what a row operation does to each of them. Recall from Theorem 2.4.7 that  $A$  is row equivalent to  $B$  exactly when  $B = EA$  with  $E \in M_m(F)$  invertible; since row equivalence is symmetric (Theorem 2.2.2 (b)), we may simply say that  $A$  and  $B$  are row equivalent.

**Theorem 2.5.1 — Row operations and the three spaces**

Let  $A, B \in M_{m \times n}(F)$  be row equivalent. Then:

- (a)  $\text{row}(B) = \text{row}(A)$ ;
- (b)  $\text{nul}(B) = \text{nul}(A)$ . Equivalently, writing  $\mathbf{b}_1, \dots, \mathbf{b}_n$  for the columns of  $B$ : for all  $c_1, \dots, c_n \in F$ , the columns of  $A$  satisfy  $c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = \mathbf{0}$  if and only if the columns of  $B$  satisfy  $c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n = \mathbf{0}$ ;
- (c) but  $\text{col}(B)$  and  $\text{col}(A)$  **need not** be equal.

*Proof.*

- (a) Write  $B = EA$  with  $E$  invertible, by Theorem 2.4.7. By Theorem 0.9.2, row  $i$  of  $EA$  is (row  $i$  of  $E$ ) $A$ , which by Theorem 0.9.3 (rows version) is a combination of the rows of  $A$ . So every row of  $B$  lies in  $\text{row}(A)$ , and  $\text{row}(B) \subseteq \text{row}(A)$  by Proposition 1.3.4 (1). Since  $A = E^{-1}B$  with  $E^{-1}$  invertible, the same argument gives  $\text{row}(A) \subseteq \text{row}(B)$ .
- (b) The equality  $\text{nul}(B) = \text{nul}(A)$  is Lemma 2.3.1. By Theorem 0.9.3,  $c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = A\mathbf{c}$  and  $c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n = B\mathbf{c}$  with  $\mathbf{c} = (c_1, \dots, c_n)$ , so the second statement is the first one written out (this is Lemma 2.3.2).
- (c) Let  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$  and  $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , obtained by  $R_2 \rightarrow R_2 - R_1$ . Then  $\text{col}(A) = \text{span}((1, 1))$  and  $\text{col}(B) = \text{span}((1, 0))$ . These lines differ, since  $(1, 0)$  is not a multiple of  $(1, 1)$ .

So row reduction keeps the row space and the **relations among the columns**, but it moves the columns themselves.

*Warning. Do not read  $\text{col}(A)$  off the reduced form.* In (c) above, the RREF of  $A$  is  $B$ , whose column space is the  $x$ -axis, while  $\text{col}(A)$  is the line  $y = x$ . What survives row reduction is the *pattern* of dependence: in both matrices column 2 equals column 1. So we use the RREF to decide **which** columns to take, and then take those columns from  $A$ .

## 2.5.4 Bases from the reduced form

Fix the notation for the rest of the section. Let  $R = (r_{kj})$  be the reduced row echelon form of  $A$  (Theorem 2.2.3, Theorem 2.3.4), with columns  $\mathbf{r}_1, \dots, \mathbf{r}_n \in F^m$ . Let  $R$  have  $r$  non-zero rows, and let the pivot of row  $k$  lie in column  $j_k$ , so that  $j_1 < j_2 < \dots < j_r$ . These are the **pivot columns** of  $A$ , well defined by Corollary 2.3.5. The remaining columns are the **free columns**. We use three facts from Definition 2.2.4:

- rows  $r + 1, \dots, m$  of  $R$  are zero;
- column  $j_k$  of  $R$  is  $\mathbf{e}_k$  (the pivot is 1 and the only non-zero entry of its column);
- in row  $k$ , every entry to the left of the pivot is 0.

From the last two, if column  $j$  of  $R$  has exactly  $t$  pivot columns to its left, then **every entry of  $\mathbf{r}_j$  in rows  $t + 1, \dots, m$  is zero, except the pivot entry of row  $t + 1$  when  $j = j_{t+1}$** . Indeed, for  $k \geq t + 1$ , row  $k$  is either zero or has its pivot in column  $j_k \geq j$ , and entries to the left of a pivot vanish. In the running example,

$$R = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

with  $r = 3$ , pivot columns 1, 3, 4 and free columns 2, 5 (the reduction is carried out in [Example \(Bases for the running example\)](#) below).

The column space comes first. In  $R$  the answer is visible: the pivot columns are  $\mathbf{e}_1, \dots, \mathbf{e}_r$ , and every column has zeros below row  $r$ . Row reduction does not preserve columns, but by [Theorem 2.5.1](#) (b) it preserves every linear relation among them, and “is a basis” and “equals this combination” are statements about relations.

### Theorem 2.5.2 — Pivot columns form a basis of the column space

Let  $A \in M_{m \times n}(F)$  have pivot columns  $j_1 < \dots < j_r$ , and let  $R = (r_{kj})$  be its RREF. Then:

- the pivot columns of  $A$ ,  $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$ , form a basis of  $\text{col}(A)$ ;
- if column  $j$  is free and  $t$  pivot columns lie to its left, then  $\mathbf{a}_j = r_{1j}\mathbf{a}_{j_1} + \dots + r_{tj}\mathbf{a}_{j_t}$ ;

- (c) the pivot columns of  $A$  are exactly the vectors kept when  $(\mathbf{a}_1, \dots, \mathbf{a}_n)$  is sifted as in Theorem 1.5.4.

*Idea.* Both claims are statements about linear relations among columns, so by Theorem 2.5.1 (b) it suffices to prove them for  $R$ , where they can be seen by inspection. The pivot columns of  $R$  are  $\mathbf{e}_1, \dots, \mathbf{e}_r$ , which are independent. A free column of  $R$  is  $\sum_k r_{kj} \mathbf{e}_k$ , a combination of the pivot columns to its left with its own entries as coefficients. So we transfer two relations back to  $A$ : “ $\sum c_k \mathbf{a}_{j_k} = \mathbf{0}$  forces  $c_1 = \dots = c_r = 0$ ” and “ $\mathbf{a}_j - \sum_k r_{kj} \mathbf{a}_{j_k} = \mathbf{0}$ ”. Part (c) is a rereading of Corollary 2.3.5.

*Proof.* Throughout, we use Theorem 2.5.1 (b) in the form: for scalars  $c_1, \dots, c_n$ ,

$$c_1 \mathbf{a}_1 + \dots + c_n \mathbf{a}_n = \mathbf{0} \iff c_1 \mathbf{r}_1 + \dots + c_n \mathbf{r}_n = \mathbf{0}. \quad (\dagger)$$

- (b) Let column  $j$  be free, with  $t$  pivot columns  $j_1, \dots, j_t$  to its left. By the fact in bold before the theorem, the entries of  $\mathbf{r}_j$  in rows  $t+1, \dots, m$  are zero, so  $\mathbf{r}_j = r_{1j} \mathbf{e}_1 + \dots + r_{tj} \mathbf{e}_t = r_{1j} \mathbf{r}_{j_1} + \dots + r_{tj} \mathbf{r}_{j_t}$ , since  $\mathbf{r}_{j_k} = \mathbf{e}_k$ . This is a linear relation among the columns of  $R$  with coefficient 1 on  $\mathbf{r}_j$ ,  $-r_{kj}$  on  $\mathbf{r}_{j_k}$ , and 0 elsewhere. By  $(\dagger)$ , the columns of  $A$  satisfy the same relation, which is (b).
- (c) *Spanning.* Each pivot column  $\mathbf{a}_{j_k}$  lies in  $\text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$ , and each free column does by (b). Hence  $\text{col}(A) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n) \subseteq \text{span}(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_r})$  by Proposition 1.3.4 (1). The reverse inclusion holds by Proposition 1.3.4 (2), since the pivot columns are among the columns.

*Independence.* Let  $c_1 \mathbf{a}_{j_1} + \dots + c_r \mathbf{a}_{j_r} = \mathbf{0}$ . This is a relation among all the columns of  $A$  with coefficient 0 on the free columns. By  $(\dagger)$ ,  $c_1 \mathbf{r}_{j_1} + \dots + c_r \mathbf{r}_{j_r} = c_1 \mathbf{e}_1 + \dots + c_r \mathbf{e}_r = \mathbf{0}$ . The  $k$ -th entry of the left side is  $c_k$ , so  $c_1 = \dots = c_r = 0$ . Hence the pivot columns of  $A$  are linearly independent, and with spanning they form a basis of  $\text{col}(A)$  by Definition 1.5.1.

- (c) By Corollary 2.3.5 (a), column  $j$  is a pivot column if and only if  $\mathbf{a}_j \notin \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ . Sifting keeps  $\mathbf{a}_j$  exactly when  $\mathbf{a}_j$  is not in the span of the vectors kept before it, and by Theorem 1.5.4 (applied to  $(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ ) that span is  $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_{j-1})$ . Hence the kept vectors are exactly the pivot columns. This proves the theorem. ■

This is sifting (Theorem 1.5.4) done by elimination. In Chapter 1, each step “is  $\mathbf{v}_j$  in the span of the vectors kept so far?” was a separate small system. One row reduction answers all of them at once, and part (b) even gives the coefficients.

The row space and the null space are read off  $R$  directly.

### Theorem 2.5.3 — Basis of the row space

The non-zero rows of the RREF  $R$  of  $A$  form a basis of  $\text{row}(A)$ .

*Proof.* By Theorem 2.5.1 (a),  $\text{row}(A) = \text{row}(R)$ . The zero rows of  $R$  contribute nothing to a combination, so the  $r$  non-zero rows span  $\text{row}(R)$ . For independence, let  $c_1(\text{row } 1) + \dots + c_r(\text{row } r) = 0$ . Look at the entry in column  $j_k$ : since column  $j_k$  of  $R$  is  $\mathbf{e}_k$ , only

row  $k$  has a non-zero entry there, equal to 1. So that entry of the combination is  $c_k$ , and  $c_k = 0$  for each  $k$ . Hence the non-zero rows are a basis of  $\text{row}(A)$ . ■

For the null space, each free variable gets one basis vector: set that free variable to 1, the other free variables to 0, and solve for the pivot variables.

### Theorem 2.5.4 — Basis of the null space

Let  $A \in M_{m \times n}(F)$  have RREF  $R = (r_{kj})$  with pivot columns  $j_1 < \dots < j_r$ . For each free column  $f$ , define  $\mathbf{s}_f \in F^n$  by

$$(\mathbf{s}_f)_f = 1, \quad (\mathbf{s}_f)_{f'} = 0 \text{ for every other free } f', \quad (\mathbf{s}_f)_{j_k} = -r_{kf} \text{ for } k = 1, \dots, r.$$

Then the vectors  $\mathbf{s}_f$ , one for each free column  $f$ , form a basis of  $\text{nul}(A)$ . In particular  $\text{nullity } A = n - r$ , the number of free columns.

*Proof.* By Theorem 2.5.1 (b),  $\text{nul}(A) = \text{nul}(R)$ . For  $k \leq r$ , row  $k$  of  $R$  has entry 1 in column  $j_k$ , entry 0 in every other pivot column (since column  $j_{k'}$  is  $\mathbf{e}_{k'}$ ), and entries  $r_{kf}$  in the free columns. Rows  $k > r$  are zero. Hence  $R\mathbf{x} = \mathbf{0}$  is equivalent to the  $r$  equations

$$x_{j_k} = - \sum_{f \text{ free}} r_{kf} x_f \quad (k = 1, \dots, r). \quad (\ddagger)$$

*Spanning.* Let  $\mathbf{x} \in \text{nul}(A)$ , and put  $\mathbf{y} = \sum_{f \text{ free}} x_f \mathbf{s}_f$ . For a free column  $f$ , the  $f$ -th entry of  $\mathbf{y}$  is  $x_f$ , since  $\mathbf{s}_f$  has a 1 there and the other  $\mathbf{s}_{f'}$  have 0. For a pivot column, the  $j_k$ -th entry of  $\mathbf{y}$  is  $\sum_f x_f (-r_{kf})$ , which equals  $x_{j_k}$  by  $(\ddagger)$ . So  $\mathbf{x} = \mathbf{y}$  lies in the span of the  $\mathbf{s}_f$ . Each  $\mathbf{s}_f$  itself satisfies  $(\ddagger)$ , so it lies in  $\text{nul}(A)$ .

*Independence.* If  $\sum_f c_f \mathbf{s}_f = \mathbf{0}$ , then the  $f$ -th entry of the left side is  $c_f$ , so every  $c_f = 0$ .

Hence the  $\mathbf{s}_f$  form a basis of  $\text{nul}(A)$ , and there are  $n - r$  of them. ■

### Example. (Bases for the running example)

Find bases of  $\text{col}(A)$ ,  $\text{row}(A)$  and  $\text{nul}(A)$ , and the rank and nullity, for

$$A = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \in M_{4 \times 5}(\mathbb{R}).$$

*Solution.* Row reduce:

$$A \xrightarrow[\substack{R_3 \rightarrow R_3 - R_1 \\ R_2 \rightarrow R_2 - 2R_1}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2}} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 & 3 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_2 \\ R_3 \rightarrow R_3 - R_2}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2}} \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 - R_3} R = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 3, 4, and the free columns are 2, 5.

**Column space.** By Theorem 2.5.2 (a), the pivot columns of  $A$ ,

$$\mathbf{a}_1 = (1, 2, 1, 0), \quad \mathbf{a}_3 = (0, 1, 1, 1), \quad \mathbf{a}_4 = (0, 0, 1, 1),$$

form a basis of  $\text{col}(A)$ , so  $\text{rank } A = 3$ . By part (b), the free columns are  $\mathbf{a}_2 = 2\mathbf{a}_1$  (from column 2 of  $R$ ) and  $\mathbf{a}_5 = -\mathbf{a}_1 + 2\mathbf{a}_3 + \mathbf{a}_4$  (from column 5). Check:  $-(1, 2, 1, 0) + 2(0, 1, 1, 1) + (0, 0, 1, 1) = (-1, 0, 2, 3) = \mathbf{a}_5$ . Note that  $\text{col}(A) \neq \text{col}(R)$ : every column of  $R$  has last entry 0, but  $\mathbf{a}_3$  does not.

**Row space.** By Theorem 2.5.3, a basis is

$$(1 \ 2 \ 0 \ 0 \ -1), \quad (0 \ 0 \ 1 \ 0 \ 2), \quad (0 \ 0 \ 0 \ 1 \ 1),$$

so  $\dim \text{row}(A) = 3$ .

**Null space.** The equations ( $\dagger$ ) read  $x_1 = -2x_2 + x_5$ ,  $x_3 = -2x_5$ ,  $x_4 = -x_5$ . By Theorem 2.5.4, a basis is

$$\mathbf{s}_2 = (-2, 1, 0, 0, 0), \quad \mathbf{s}_5 = (1, 0, -2, -1, 1),$$

so  $\text{nullity } A = 2$ . Check:  $A\mathbf{s}_2 = -2\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{0}$  and  $A\mathbf{s}_5 = \mathbf{a}_1 - 2\mathbf{a}_3 - \mathbf{a}_4 + \mathbf{a}_5 = \mathbf{0}$ , which are the two relations found above.

So  $\text{rank } A = \dim \text{row}(A) = 3$  and  $\text{rank } A + \text{nullity } A = 5$ , the number of columns.

*Quick check.* Without computing anything, what is the largest possible rank of a  $3 \times 5$  matrix, and the smallest possible nullity?

*Solution.*  $\text{rank } A = \dim \text{col}(A) \leq 3$ , since  $\text{col}(A) \subseteq F^3$ . By Theorem 2.5.4, the nullity is the number of free columns,  $5 - r$ , where  $r \leq 3$  is the number of pivots (one per non-zero row). So the nullity is at least 2. For example  $(I_3 \ 0_{3 \times 2})$  attains rank 3 and nullity 2.

## 2.5.5 Row rank equals column rank

The example came out with  $\dim \text{row}(A) = \dim \text{col}(A)$ . This is not luck, and it is surprising: the row space and the column space live in different spaces, and the rows and columns of a  $4 \times 5$  matrix have nothing obvious in common. Counting pivots explains it.

### Theorem 2.5.5 — Row rank equals column rank

Let  $A \in M_{m \times n}(F)$ , and let  $r$  be the number of pivots of its RREF. Then

$$\dim \text{row}(A) = \dim \text{col}(A) = r.$$

Consequently  $\text{rank } A = \text{rank } A^T$ .

*Proof.* By Theorem 2.5.2 (a),  $\text{col}(A)$  has a basis consisting of the  $r$  pivot columns, so  $\dim \text{col}(A) = r$ . By Theorem 2.5.3,  $\text{row}(A)$  has a basis consisting of the non-zero rows of the RREF. Each non-zero row contains exactly one pivot and each pivot lies in a

non-zero row, so there are  $r$  of them, and  $\dim \text{row}(A) = r$ . Finally, by (\*) at the start of this section,  $\text{rank } A^T = \dim \text{col}(A^T) = \dim \text{row}(A) = r = \text{rank } A$ .

■

The theorem makes the rank independent of **which** echelon form we compute. Gaussian elimination without the final clearing-up phase produces an echelon form that is not reduced, and different sequences of operations produce different echelon forms. They all give the same count.

### Corollary 2.5.6 — Rank from any echelon form

Let  $A \in M_{m \times n}(F)$ , and let  $G$  be a matrix in row echelon form that is row equivalent to  $A$ . Then the number of non-zero rows of  $G$  is  $\text{rank } A$ , and the columns of  $G$  containing its pivots are exactly the pivot columns of  $A$ .

*Proof.* By Corollary 2.3.5 (b), the pivot columns of  $G$  are the pivot columns of  $A$ . Each non-zero row of  $G$  has exactly one pivot, in a different column (Definition 2.2.3), so the number of non-zero rows of  $G$  equals the number of pivot columns of  $A$ , which is  $\text{rank } A$  by Theorem 2.5.5.

■

*Warning.* Rank is a property of  $A$ , not of a computation, but it must be counted from a row echelon form. The number of non-zero rows of  $A$  itself is not the rank:  $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$  has two non-zero rows and rank 1. Count non-zero rows only after reaching an echelon form. And the rank depends on the field when the entries are read in different fields:  $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  has rank 2 over  $\mathbb{R}$ , but over  $\mathbb{F}_2$ , where  $-1 = 1$ , both rows are  $(1 \ 1)$  and the rank is 1.

## 2.5.6 Rank–nullity and the Rouché–Capelli theorem

The trade-off between the zero matrix and the identity was exact, and now we can say why.

### Theorem 2.5.7 — Rank–Nullity for Matrices

For every  $A \in M_{m \times n}(F)$ ,

$$\text{rank } A + \text{nullity } A = n,$$

the number of **columns** of  $A$ .

*Proof.* Let  $r$  be the number of pivot columns of  $A$ . By Theorem 2.5.5,  $\text{rank } A = r$ . By Theorem 2.5.4,  $\text{nullity } A = n - r$ . Adding gives  $n$ .

■

Each column of  $A$  is either a pivot column, contributing a direction to  $\text{col}(A)$ , or a free column, contributing a direction to  $\text{nul}(A)$ . Nothing is lost and nothing is counted twice. In Chapter 3 the same equation becomes the Rank–Nullity Theorem for linear maps, with a proof that uses no matrices at all.

Now the question this chapter started with. When does  $A\mathbf{x} = \mathbf{b}$  have a solution, and how many solutions are there? The first answer was  $\mathbf{b} \in \text{col}(A)$  (Theorem 2.1.1); here is a version that is a pure count.

### Theorem 2.5.8 — Rouché–Capelli Theorem

Let  $A \in M_{m \times n}(F)$ ,  $\mathbf{b} \in F^m$ , and let  $[A \mid \mathbf{b}] \in M_{m \times (n+1)}(F)$  be the augmented matrix.

- (a) The system  $A\mathbf{x} = \mathbf{b}$  is consistent if and only if  $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$ .
- (b) Suppose it is consistent, let  $\mathbf{p}$  be one solution, let  $r = \text{rank } A$ , and let  $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$  be a basis of  $\text{nul}(A)$ . Then every solution can be written as

$$\mathbf{x} = \mathbf{p} + t_1\mathbf{s}_1 + \cdots + t_{n-r}\mathbf{s}_{n-r}$$

for **exactly one** choice of  $(t_1, \dots, t_{n-r}) \in F^{n-r}$ . In words: the solutions form an  $(n-r)$ -parameter family.

- (c) If moreover  $F$  is a finite field with  $q$  elements (for instance  $\mathbb{F}_p$ , with  $q = p$ ), a consistent system has exactly  $q^{n-\text{rank } A}$  solutions.

*Idea.* For (a), adding the column  $\mathbf{b}$  can only enlarge the column space, and the dimension goes up exactly when  $\mathbf{b}$  is new, that is, when  $\mathbf{b} \notin \text{col}(A)$ . For (b), the structure theorem says solutions are  $\mathbf{p}$  plus null vectors, and null vectors have unique coordinates in a basis. For (c), unique coordinates mean the solutions are in bijection with  $F^{n-r}$ , which has  $q^{n-r}$  elements.

*Proof.*

- (a) By Definition 2.5.1,  $\text{col}([A \mid \mathbf{b}]) = \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b})$ , which contains  $\text{col}(A)$  by Proposition 1.3.4 (2).  
 $(\Rightarrow)$  Suppose the system is consistent. By Theorem 2.1.1,  $\mathbf{b} \in \text{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ , so by Theorem 1.3.5  $\text{col}([A \mid \mathbf{b}]) = \text{col}(A)$ , and the ranks are equal.  
 $(\Leftarrow)$  Suppose  $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$ . Then  $\text{col}(A)$  is a subspace of  $\text{col}([A \mid \mathbf{b}])$  of the same dimension, so  $\text{col}(A) = \text{col}([A \mid \mathbf{b}])$  by Theorem 1.6.8. In particular  $\mathbf{b} \in \text{col}(A)$ , and the system is consistent by Theorem 2.1.1.
- (b) By Theorem 2.5.7, nullity  $A = n - r$ , so a basis  $\mathbf{s}_1, \dots, \mathbf{s}_{n-r}$  of  $\text{nul}(A)$  exists (Theorem 2.5.4). By Theorem 2.1.3,  $\mathbf{x}$  is a solution if and only if  $\mathbf{x} - \mathbf{p} \in \text{nul}(A)$ . By Theorem 1.5.1, applied in the vector space  $\text{nul}(A)$  with this basis, each  $\mathbf{x} - \mathbf{p} \in \text{nul}(A)$  equals  $t_1\mathbf{s}_1 + \cdots + t_{n-r}\mathbf{s}_{n-r}$  for exactly one  $(t_1, \dots, t_{n-r})$ . Conversely every such expression is in  $\text{nul}(A)$ , so  $\mathbf{p}$  plus it is a solution. This proves (b).
- (c) By (b), the rule  $(t_1, \dots, t_{n-r}) \mapsto \mathbf{p} + t_1\mathbf{s}_1 + \cdots + t_{n-r}\mathbf{s}_{n-r}$  is a bijection from  $F^{n-r}$  onto the solution set: every solution is hit, and by exactly one tuple. A tuple in  $F^{n-r}$  is a choice of  $n-r$  entries, each one of  $q$  elements, so  $F^{n-r}$  has  $q^{n-r}$  elements. (When  $r = n$  there is exactly one tuple, the empty one, and the basis of  $\text{nul}(A)$  is empty, so  $\mathbf{p}$  is the only solution and  $q^0 = 1$ .) Hence the solution set has  $q^{n-r}$  elements.

Over an infinite field such as  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\mathbb{C}$ , part (b) gives the familiar trichotomy: no solution, exactly one solution (when  $\text{rank } A = n$ ), or infinitely many (when  $\text{rank } A < n$ , since distinct values of  $t_1$  give distinct solutions). Over a finite field “infinitely many” is replaced by an exact power of  $q$ . For example, over  $\mathbb{F}_2$  a system can never have exactly 3 or 6 solutions.

**Example. (Counting solutions over  $\mathbb{F}_3$ )**

Over  $\mathbb{F}_3$ , consider

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 2 & 0 & 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{b}' = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Decide whether  $A\mathbf{x} = \mathbf{b}$  and  $A\mathbf{x} = \mathbf{b}'$  are consistent, and count their solutions.

*Solution.* We reduce both augmented matrices at once, carrying two right-hand columns; arithmetic is modulo 3, so  $-1 = 2$  and  $-2 = 1$ .

$$\left[ \begin{array}{cccc|c|c} 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{cccc|c|c} 1 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 1 \end{array} \right] \xrightarrow[R_1 \rightarrow R_1 - R_2]{R_3 \rightarrow R_3 - R_2} \left[ \begin{array}{cccc|c|c} 1 & 0 & 2 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{array} \right].$$

Row operations act on each column separately, so deleting either right-hand column gives a reduction of the corresponding augmented matrix. For instance, in row 2,  $(1, 2, 1, 0) - (1, 1, 0, 2) = (0, 1, 1, -2) = (0, 1, 1, 1)$ , and in row 3,  $(2, 0, 1, 2) - 2(1, 1, 0, 2) = (0, -2, 1, -2) = (0, 1, 1, 1)$ .

The left block is an echelon form of  $A$  with two non-zero rows, so  $\text{rank } A = 2$  by [Corollary 2.5.6](#). With  $\mathbf{b}$ , the augmented matrix  $[A \mid \mathbf{b}]$  has the same two non-zero rows (the last is zero), so  $\text{rank } [A \mid \mathbf{b}] = 2 = \text{rank } A$ , and by [Theorem 2.5.8](#) (a) the system is consistent. By part (c) it has exactly  $3^{4-2} = 9$  solutions. Explicitly, the reduced system is  $x_1 = 2 - 2x_3 - x_4 = 2 + x_3 + 2x_4$  and  $x_2 = 2 - x_3 - x_4 = 2 + 2x_3 + 2x_4$ , with  $x_3, x_4 \in \mathbb{F}_3$  free: nine choices.

With  $\mathbf{b}'$ , the last row  $(0, 0, 0, 0 \mid 2)$  is a non-zero row with its pivot in the last column, so  $[A \mid \mathbf{b}']$  has three non-zero rows in echelon form and  $\text{rank } [A \mid \mathbf{b}'] = 3 \neq 2$ . By [Theorem 2.5.8](#) (a),  $A\mathbf{x} = \mathbf{b}'$  has no solution. Indeed, row 3 of  $A$  is the sum of rows 1 and 2, so any solution would need  $b'_3 = b'_1 + b'_2 = 1$ , but  $b'_3 = 0$ .

## 2.5.7 Rank of a product

Multiplying matrices cannot create information: every column of  $AB$  is a combination of the columns of  $A$ , and every row is a combination of the rows of  $B$ . The rank records this.

### Theorem 2.5.9 — Rank of a product

Let  $A \in M_{m \times n}(F)$  and  $B \in M_{n \times p}(F)$ . Then

$$\text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B).$$

Moreover, if  $P \in M_m(F)$  and  $Q \in M_n(F)$  are invertible, then  $\text{rank}(PA) = \text{rank } A = \text{rank}(AQ)$ .

*Proof.* Let  $\mathbf{b}_1, \dots, \mathbf{b}_p \in F^n$  be the columns of  $B$ . By Theorem 0.9.2, the columns of  $AB$  are  $A\mathbf{b}_1, \dots, A\mathbf{b}_p$ , which lie in  $\text{col}(A)$  by Definition 2.5.1. By Proposition 1.3.4 (1),  $\text{col}(AB) \subseteq \text{col}(A)$ , so  $\text{rank}(AB) \leq \text{rank } A$  by Theorem 1.6.7. Similarly, row  $i$  of  $AB$  is (row  $i$  of  $A$ ) $B$ , a combination of the rows of  $B$  by Theorem 0.9.3 (rows version), so  $\text{row}(AB) \subseteq \text{row}(B)$ . By Theorem 2.5.5,  $\text{rank}(AB) = \dim \text{row}(AB) \leq \dim \text{row}(B) = \text{rank } B$ .

For the second statement, apply the inequality twice:  $\text{rank } A = \text{rank}(P^{-1}(PA)) \leq \text{rank}(PA) \leq \text{rank } A$ , so equality holds throughout. The same sandwich with  $A = (AQ)Q^{-1}$  gives  $\text{rank}(AQ) = \text{rank } A$ . ■

The equality case recovers something we already knew in a new form: row operations (left multiplication by an invertible  $P$ ) do not change the rank. The inequality is often strict. For  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , both factors of  $A \cdot A$  have rank 1, but  $A^2 = 0$  has rank 0.

## 2.5.8 A toolkit for questions from Chapter 1

In Chapter 1 we asked, again and again, whether some vectors in  $F^m$  are independent, whether they span, how to extend them to a basis, and what the sum and intersection of two subspaces are. Each was answered by solving a small system by hand. Everything now reduces to one move: **put the vectors as columns of a matrix, row reduce, and read off pivots.**

### Theorem 2.5.10 — Independence and spanning by rank

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^m$ , and let  $A = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_k) \in M_{m \times k}(F)$ . Then:

- (a)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is linearly independent if and only if  $\text{rank } A = k$ , that is, every column is a pivot column;
- (b)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  spans  $F^m$  if and only if  $\text{rank } A = m$ , that is, every row of the RREF has a pivot;
- (c)  $(\mathbf{v}_1, \dots, \mathbf{v}_k)$  is a basis of  $F^m$  if and only if  $k = m = \text{rank } A$ .

*Proof.*

- (a) By Theorem 0.9.3,  $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = A\mathbf{c}$  with  $\mathbf{c} = (c_1, \dots, c_k)$ , so the list is independent if and only if  $\text{nul}(A) = \{\mathbf{0}\}$ , that is, nullity  $A = 0$ . By Theorem 2.5.7 this means  $\text{rank } A = k$ .
- (b) The list spans  $F^m$  if and only if  $\text{col}(A) = F^m$ . Since  $\text{col}(A) \subseteq F^m$  and  $\dim F^m = m$ , this holds if and only if  $\text{rank } A = m$ , by Theorem 1.6.8 for one direction and because

equal spaces have equal dimension for the other. The number of pivots is rank  $A$  by [Theorem 2.5.5](#), and the RREF has  $m$  rows, each with at most one pivot.

(c) Combine (a) and (b) with Definition 1.5.1.

■

### Example. (Testing independence)

Decide whether  $\mathbf{v}_1 = (1, 2, 0, 1)$ ,  $\mathbf{v}_2 = (2, 4, 1, 1)$ ,  $\mathbf{v}_3 = (0, 0, 1, -1)$  are linearly independent in  $\mathbb{R}^4$ . If not, find a linear relation.

*Solution.* Row reduce the matrix with these columns:

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_1 \\ R_2 \rightarrow R_2 - 2R_1}]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow[\substack{R_1 \rightarrow R_1 - 2R_2 \\ R_4 \rightarrow R_4 + R_2}]{R_4 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Only columns 1 and 2 are pivot columns, so the rank is  $2 < 3$ , and the vectors are **dependent** by [Theorem 2.5.10](#) (a). Column 3 of the RREF gives the relation, by [Theorem 2.5.2](#) (b):  $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$ . Check:  $-2(1, 2, 0, 1) + (2, 4, 1, 1) = (0, 0, 1, -1)$ . Moreover  $(\mathbf{v}_1, \mathbf{v}_2)$  is a basis of  $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  by [Theorem 2.5.2](#) (a).

To **extend** an independent list to a basis of  $F^m$ , append the standard basis and keep the pivot columns. The appended vectors guarantee that the columns span  $F^m$ . The original vectors all survive, because by [Theorem 2.5.2](#) (c) a column is discarded only if it lies in the span of the columns before it, and an independent list has no such vector (Lemma 1.5.3).

### Example. (Extending to a basis by elimination)

Extend the independent list  $(\mathbf{v}_1, \mathbf{v}_2) = ((1, 2, 0, 1), (2, 4, 1, 1))$  from [Example \(Testing independence\)](#) to a basis of  $\mathbb{R}^4$ .

*Solution.* Form  $M = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$  and row reduce:

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_1 \\ R_2 \rightarrow R_2 - 2R_1}]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[\text{then } R_4 \rightarrow R_4 + R_2]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 1 \end{pmatrix}$$

$$\xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & -2 & 1 & 0 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & -2 \end{pmatrix}.$$

This is an echelon form with pivots in columns 1, 2, 3, 4. By [Corollary 2.5.6](#) these are the pivot columns of  $M$ , so by [Theorem 2.5.2](#) (a) the corresponding columns of  $M$  form a basis of  $\text{col}(M) = \mathbb{R}^4$ :

$$(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2).$$

(Clearing above the pivots gives the RREF, whose column 5 says  $\mathbf{e}_3 = -\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{e}_1 - 2\mathbf{e}_2$ ; check:  $-(1, 2, 0, 1) + (2, 4, 1, 1) - (1, 0, 0, 0) - 2(0, 1, 0, 0) = (0, 0, 1, 0)$ .)

For sums and intersections, one reduction does both jobs.

### Proposition 2.5.11 — Sum and intersection by elimination

Let  $U, W \subseteq F^m$  be subspaces with bases  $(\mathbf{u}_1, \dots, \mathbf{u}_k)$  and  $(\mathbf{w}_1, \dots, \mathbf{w}_l)$ , and let  $M = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \ \mathbf{w}_1 \ \cdots \ \mathbf{w}_l) \in M_{m \times (k+l)}(F)$ . Then:

- (a) the pivot columns of  $M$  form a basis of  $U + W$ ;
- (b) for  $\mathbf{x} = (x_1, \dots, x_{k+l}) \in F^{k+l}$  write  $\mathbf{u}_{\mathbf{x}} := x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k$ , the combination of the  $\mathbf{u}$ 's with the first  $k$  entries of  $\mathbf{x}$  as coefficients. If  $(\mathbf{n}_1, \dots, \mathbf{n}_d)$  is a basis of  $\text{nul}(M)$ , then  $(\mathbf{u}_{\mathbf{n}_1}, \dots, \mathbf{u}_{\mathbf{n}_d})$  is a basis of  $U \cap W$ .

In particular  $\dim(U + W) = \text{rank } M$  and  $\dim(U \cap W) = \text{nullity } M$ .

*Proof.*

- (a) By Proposition 1.7.2,  $U + W = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_l) = \text{col}(M)$ , and Theorem 2.5.2 (a) applies.
- (b) By Theorem 0.9.3,  $\mathbf{x} \in \text{nul}(M)$  means  $x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k = -(x_{k+1}\mathbf{w}_1 + \cdots + x_{k+l}\mathbf{w}_l)$ , so  $\mathbf{u}_{\mathbf{x}}$  lies in  $U$  and in  $W$ . Also  $\mathbf{u}_{\mathbf{x}+\mathbf{y}} = \mathbf{u}_{\mathbf{x}} + \mathbf{u}_{\mathbf{y}}$  and  $\mathbf{u}_{c\mathbf{x}} = c\mathbf{u}_{\mathbf{x}}$ , by the vector space axioms in  $F^m$ .

*Spanning.* Let  $\mathbf{v} \in U \cap W$ . Since the  $\mathbf{u}$ 's span  $U$  and the  $\mathbf{w}$ 's span  $W$ ,  $\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = b_1\mathbf{w}_1 + \cdots + b_l\mathbf{w}_l$ . Then  $\mathbf{x} = (a_1, \dots, a_k, -b_1, \dots, -b_l) \in \text{nul}(M)$  and  $\mathbf{u}_{\mathbf{x}} = \mathbf{v}$ . Writing  $\mathbf{x} = \sum_i t_i \mathbf{n}_i$  gives  $\mathbf{v} = \sum_i t_i \mathbf{u}_{\mathbf{n}_i}$ .

*Independence.* Let  $\sum_i t_i \mathbf{u}_{\mathbf{n}_i} = \mathbf{0}$ , and put  $\mathbf{x} = \sum_i t_i \mathbf{n}_i \in \text{nul}(M)$ . Then  $\mathbf{u}_{\mathbf{x}} = x_1\mathbf{u}_1 + \cdots + x_k\mathbf{u}_k = \mathbf{0}$ , so  $x_1 = \cdots = x_k = 0$  because the  $\mathbf{u}$ 's are independent. Since  $M\mathbf{x} = \mathbf{0}$ , also  $x_{k+1}\mathbf{w}_1 + \cdots + x_{k+l}\mathbf{w}_l = \mathbf{0}$ , so the remaining entries vanish because the  $\mathbf{w}$ 's are independent. Hence  $\mathbf{x} = \mathbf{0}$ , and  $t_1 = \cdots = t_d = 0$  because the  $\mathbf{n}_i$  are independent.

The dimension statements follow by counting the two bases, using Theorem 2.5.5 for rank  $M$ . ■

Adding the two counts and using Theorem 2.5.7 gives  $\dim(U + W) + \dim(U \cap W) = \text{rank } M + \text{nullity } M = k + l = \dim U + \dim W$ . This is the Dimension Formula for Sums (Theorem 1.7.3) again, now with a second, computational proof for subspaces of  $F^m$ . (Some texts use the matrix with columns  $\mathbf{u}_1, \dots, \mathbf{u}_k, -\mathbf{w}_1, \dots, -\mathbf{w}_l$  instead of  $M$ , so that a null vector gives equal combinations  $\sum a_i \mathbf{u}_i = \sum b_j \mathbf{w}_j$  with no sign change. The two null spaces differ only by the signs of the last  $l$  entries.)

#### Example. (A basis of a sum)

Let  $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2)$  and  $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$  in  $\mathbb{R}^4$ , where

$$\mathbf{u}_1 = (1, 0, 1, 2), \quad \mathbf{u}_2 = (0, 1, 1, 1), \quad \mathbf{w}_1 = (1, 1, 0, 1), \quad \mathbf{w}_2 = (1, 3, 0, 1).$$

Find a basis and the dimension of  $U + W$ .

*Solution.* Each pair is independent, since neither vector is a multiple of the other (compare the first two entries), so  $\dim U = \dim W = 2$ . Row reduce  $M = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{w}_1 \ \mathbf{w}_2)$ :

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - 2R_1}]{\substack{R_3 \rightarrow R_3 - R_1}} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{pmatrix} \xrightarrow[\substack{R_4 \rightarrow R_4 - R_2}]{\substack{R_3 \rightarrow R_3 - R_2}} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -2 & -4 \end{pmatrix} \\ \xrightarrow[\substack{\text{then } R_3 \rightarrow -\frac{1}{2}R_3}]{\substack{R_4 \rightarrow R_4 - R_3}} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 - R_3}]{\substack{R_1 \rightarrow R_1 - R_3}} R = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 2, 3. By [Proposition 2.5.11](#) (a),  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{w}_1)$  is a basis of  $U + W$ , and  $\dim(U + W) = 3$ . Column 4 of  $R$  says  $\mathbf{w}_2 = -\mathbf{u}_1 + \mathbf{u}_2 + 2\mathbf{w}_1$ ; check:  $-(1, 0, 1, 2) + (0, 1, 1, 1) + 2(1, 1, 0, 1) = (1, 3, 0, 1)$ .

### Example. (A basis of an intersection)

For  $U$  and  $W$  as in [Example \(A basis of a sum\)](#), find a basis of  $U \cap W$ , and check the Dimension Formula for Sums.

*Solution.* We reuse the RREF  $R$  of  $M$ . Its only free column is column 4, so by [Theorem 2.5.4](#)  $\text{nul}(M)$  has the basis  $\mathbf{n} = (1, -1, -2, 1)$  (set  $x_4 = 1$ ; then  $x_1 = 1$ ,  $x_2 = -1$ ,  $x_3 = -2$ ). The relation  $M\mathbf{n} = \mathbf{0}$  reads  $\mathbf{u}_1 - \mathbf{u}_2 - 2\mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$ , that is,

$$\mathbf{u}_1 - \mathbf{u}_2 = 2\mathbf{w}_1 - \mathbf{w}_2.$$

By [Proposition 2.5.11](#) (b),  $\mathbf{u}_{\mathbf{n}} = \mathbf{u}_1 - \mathbf{u}_2 = (1, -1, 0, 1)$  is a basis of  $U \cap W$ . Check that it lies in  $W$ :  $2(1, 1, 0, 1) - (1, 3, 0, 1) = (1, -1, 0, 1)$ . So  $\dim(U \cap W) = 1$ , and

$$\dim U + \dim W - \dim(U \cap W) = 2 + 2 - 1 = 3 = \dim(U + W),$$

in agreement with Theorem 1.7.3.

## 2.5.9 Exercises

### 2.5.9.1 A. Check your understanding

#### Exercise A1.

- Define  $\text{col}(A)$ ,  $\text{row}(A)$ ,  $\text{nul}(A)$  and  $\text{rank } A$  for  $A \in M_{m \times n}(F)$ , and say which space each subspace lives in.
- State the Rank–Nullity Theorem for matrices and the Rouché–Capelli Theorem.
- True or false: if  $R$  is the RREF of  $A$ , then  $\text{col}(R) = \text{col}(A)$ . Justify your answer.
- True or false: there is a  $4 \times 6$  real matrix with nullity 1. Justify your answer.
- True or false: for every  $A \in M_{m \times n}(F)$ ,  $\text{rank } A = \text{rank } A^T$ . Justify your answer.

- (f) Describe how to find a basis of  $U \cap W$  for subspaces  $U, W \subseteq F^m$  given by bases.

*Solution.*

- (a)  $\text{col}(A)$  is the span of the columns, a subspace of  $F^m$ ;  $\text{row}(A)$  is the span of the rows, a subspace of  $M_{1 \times n}(F)$ ;  $\text{nul}(A) = \{\mathbf{x} \in F^n : A\mathbf{x} = \mathbf{0}\}$ , a subspace of  $F^n$ ; and  $\text{rank } A = \dim \text{col}(A)$  (Definition 2.5.1, Definition 2.5.2, Definition 2.5.3, Definition 2.5.4).
- (b) For  $A \in M_{m \times n}(F)$ ,  $\text{rank } A + \text{nullity } A = n$  (Theorem 2.5.7). For  $\mathbf{b} \in F^m$ ,  $A\mathbf{x} = \mathbf{b}$  is consistent if and only if  $\text{rank } A = \text{rank } [A \mid \mathbf{b}]$ ; then the solutions are  $\mathbf{p} + \text{nul}(A)$ , an  $(n - \text{rank } A)$ -parameter family, with exactly  $q^{n - \text{rank } A}$  solutions over a field with  $q$  elements (Theorem 2.5.8).
- (c) False. For  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ,  $R = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ , and  $(1, 1) \in \text{col}(A)$  but  $(1, 1) \notin \text{col}(R) = \text{span}((1, 0))$ . Row operations preserve the row space and the column relations, not the column space (Theorem 2.5.1).
- (d) False. Its rank is at most 4, since the column space lies in  $\mathbb{R}^4$ , so by Theorem 2.5.7 its nullity is  $6 - \text{rank } A \geq 2$ .
- (e) True. This is Theorem 2.5.5.
- (f) Put the basis vectors of  $U$  and then of  $W$  as the columns of  $M$ , find a basis of  $\text{nul}(M)$  from the RREF, and for each basis vector  $\mathbf{n}$  take the combination of the  $\mathbf{u}$ 's with the first  $k$  entries of  $\mathbf{n}$  as coefficients (Proposition 2.5.11).

### 2.5.9.2 B. Practice

#### Exercise B1: The three subspaces of a $3 \times 5$ matrix.

Let

$$A = \begin{pmatrix} 1 & -1 & 2 & 0 & 3 \\ 2 & -2 & 5 & 1 & 4 \\ -1 & 1 & -1 & 1 & -5 \end{pmatrix} \in M_{3 \times 5}(\mathbb{R}).$$

Find bases of  $\text{col}(A)$ ,  $\text{row}(A)$  and  $\text{nul}(A)$ . Hence verify  $\text{rank } A + \text{nullity } A = 5$ .

*Solution.* Row reduce:

$$A \xrightarrow[\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1}]{\substack{R_3 \rightarrow R_3 - R_2 \\ R_1 \rightarrow R_1 - 2R_2}} R = \begin{pmatrix} 1 & -1 & 0 & -2 & 7 \\ 0 & 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1 and 3; the free columns are 2, 4, 5.

By Theorem 2.5.2, the pivot columns of  $A$ ,  $(1, 2, -1)$  and  $(2, 5, -1)$ , form a basis of  $\text{col}(A)$ . By Theorem 2.5.3,  $(1 \ -1 \ 0 \ -2 \ 7)$  and  $(0 \ 0 \ 1 \ 1 \ -2)$  form a basis of  $\text{row}(A)$ . The reduced equations are  $x_1 = x_2 + 2x_4 - 7x_5$  and  $x_3 = -x_4 + 2x_5$ , so by Theorem 2.5.4 a basis of  $\text{nul}(A)$  is

$$\mathbf{s}_2 = (1, 1, 0, 0, 0), \quad \mathbf{s}_4 = (2, 0, -1, 1, 0), \quad \mathbf{s}_5 = (-7, 0, 2, 0, 1).$$

For example  $A\mathbf{s}_4 = 2\mathbf{a}_1 - \mathbf{a}_3 + \mathbf{a}_4 = (2, 4, -2) - (2, 5, -1) + (0, 1, 1) = \mathbf{0}$ . Hence  $\text{rank } A = 2$  and  $\text{nullity } A = 3$ , and  $2 + 3 = 5$ , as Theorem 2.5.7 requires.

**Exercise B2: Independence and extension in  $\mathbb{R}^4$ .**

Let  $\mathbf{v}_1 = (1, 0, 2, 1)$ ,  $\mathbf{v}_2 = (2, 1, 3, 1)$ ,  $\mathbf{v}_3 = (0, 1, -1, -1)$  in  $\mathbb{R}^4$ .

- (a) Determine whether  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$  is linearly independent. Justify your answer.
- (b) Find a basis of  $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ , and extend it to a basis of  $\mathbb{R}^4$  using standard basis vectors.

*Solution.*

- (a) Row reduce the matrix with columns  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ :

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 3 & -1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 + R_2, R_1 \rightarrow R_1 - 2R_2]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The rank is  $2 < 3$ , so the list is dependent by [Theorem 2.5.10](#) (a). Column 3 gives  $\mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$  ([Theorem 2.5.2](#) (b)); check:  $(-2, 0, -4, -2) + (2, 1, 3, 1) = (0, 1, -1, -1)$ .

- (b) By [Theorem 2.5.2](#) (a),  $(\mathbf{v}_1, \mathbf{v}_2)$  is a basis of  $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ . To extend it, row reduce  $(\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{e}_1 \ \mathbf{e}_2 \ \mathbf{e}_3 \ \mathbf{e}_4)$ :

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 - R_1]{R_3 \rightarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & -2 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \rightarrow R_4 + R_2]{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow[\text{then } R_3 \rightarrow -R_3]{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & -2 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{R_4 \rightarrow R_4 + 2R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 1 & -2 \end{pmatrix}.$$

This echelon form has pivots in columns 1, 2, 3, 4, which are the pivot columns by [Corollary 2.5.6](#). By [Theorem 2.5.2](#) (a) the corresponding columns form a basis of the column space, which is  $\mathbb{R}^4$  since it contains  $\mathbf{e}_1, \dots, \mathbf{e}_4$ . Hence  $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{e}_1, \mathbf{e}_2)$  is a basis of  $\mathbb{R}^4$  extending  $(\mathbf{v}_1, \mathbf{v}_2)$ .

**Exercise B3: Sum and intersection in  $\mathbb{R}^5$ .**

Let  $U = \text{span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$  and  $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$  in  $\mathbb{R}^5$ , where

$$\mathbf{u}_1 = (1, 1, 0, 0, 1), \quad \mathbf{u}_2 = (0, 1, 1, 0, 0), \quad \mathbf{u}_3 = (1, 0, 0, 1, 0), \quad \mathbf{w}_1 = (1, 2, 2, 1, 2), \quad \mathbf{w}_2 = (0, 0, 1, 1, 1).$$

Find  $\dim U$ ,  $\dim W$ , a basis of  $U + W$  and a basis of  $U \cap W$ . Hence verify [Theorem 1.7.3](#).

*Solution.* Row reduce  $M = (\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3 \ \mathbf{w}_1 \ \mathbf{w}_2)$ :

$$\begin{aligned} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 & 1 \end{pmatrix} &\xrightarrow[\substack{R_2 \rightarrow R_2 - R_1 \\ R_5 \rightarrow R_5 - R_1}]{\substack{R_2 \rightarrow R_2 - R_1 \\ R_5 \rightarrow R_5 - R_1}} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix} \\ &\xrightarrow[\substack{R_4 \rightarrow R_4 - R_3 \\ R_5 \rightarrow R_5 + R_3}]{\substack{R_4 \rightarrow R_4 - R_3 \\ R_5 \rightarrow R_5 + R_3}} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix} \xrightarrow[\text{then } R_4 \rightarrow \frac{1}{2}R_4]{R_4 \leftrightarrow R_5} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Clearing above the pivots ( $R_3 \rightarrow R_3 - R_4$ ,  $R_2 \rightarrow R_2 - R_4$ ,  $R_1 \rightarrow R_1 - R_4$ , then  $R_2 \rightarrow R_2 + R_3$ ,  $R_1 \rightarrow R_1 - R_3$ ) gives

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns are 1, 2, 3, 4, so  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$  is independent by [Theorem 2.5.2](#) (a), and so are its sub-lists  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$  and  $(\mathbf{w}_1)$ ; also  $\mathbf{w}_2$  is not a multiple of  $\mathbf{w}_1$  (compare first entries). Hence  $\dim U = 3$  and  $\dim W = 2$ , with the given spanning lists as bases. By [Proposition 2.5.11](#) (a),  $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{w}_1)$  is a basis of  $U + W$ , so  $\dim(U + W) = 4$ .

The only free column is column 5. Setting  $x_5 = 1$  gives  $\mathbf{n} = (1, 1, 0, -1, 1)$ , a basis of  $\text{nul}(M)$  by [Theorem 2.5.4](#). So  $\mathbf{u}_1 + \mathbf{u}_2 - \mathbf{w}_1 + \mathbf{w}_2 = \mathbf{0}$ , and by [Proposition 2.5.11](#) (b),  $\mathbf{u}_1 + \mathbf{u}_2 = (1, 2, 1, 0, 1)$  is a basis of  $U \cap W$ . Check:  $\mathbf{w}_1 - \mathbf{w}_2 = (1, 2, 1, 0, 1)$ . Hence  $\dim(U \cap W) = 1$ , and

$$\dim U + \dim W - \dim(U \cap W) = 3 + 2 - 1 = 4 = \dim(U + W).$$

### 2.5.9.3 C. Going deeper

#### Exercise C1: Rank of a sum.

Let  $A, B \in M_{m \times n}(F)$ . Prove that  $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$ . Give an example where the inequality is strict.

*Solution.* The  $j$ -th column of  $A + B$  is  $\mathbf{a}_j + \mathbf{b}_j$ , which lies in  $\text{col}(A) + \text{col}(B)$  (Definition 1.7.1). This sum is a subspace (Theorem 1.7.1), so by Theorem 1.3.1,  $\text{col}(A + B) \subseteq \text{col}(A) + \text{col}(B)$ . By Theorem 1.6.7 and Theorem 1.7.3,

$$\text{rank}(A+B) \leq \dim(\text{col}(A) + \text{col}(B)) = \text{rank } A + \text{rank } B - \dim(\text{col}(A) \cap \text{col}(B)) \leq \text{rank } A + \text{rank } B.$$

For strictness, take  $A = I_2$  and  $B = -I_2$  over  $\mathbb{R}$ :  $\text{rank}(A + B) = \text{rank } 0 = 0 < 2 + 2$ .

**Exercise C2: Matrices of rank one.**

Let  $A \in M_{m \times n}(F)$ . Prove that  $\text{rank } A = 1$  if and only if  $A = \mathbf{x}\mathbf{y}^T$  for some **non-zero**  $\mathbf{x} \in F^m$  and  $\mathbf{y} \in F^n$ .

*Solution.* ( $\Leftarrow$ ) Suppose  $A = \mathbf{x}\mathbf{y}^T$  with  $\mathbf{x}, \mathbf{y} \neq \mathbf{0}$ , and write  $\mathbf{y} = (y_1, \dots, y_n)$ . By Theorem 0.9.2, the  $j$ -th column of  $\mathbf{x}\mathbf{y}^T$  is  $\mathbf{x}$  times the  $j$ -th column of  $\mathbf{y}^T$ , the  $1 \times 1$  matrix  $(y_j)$ ; that is,  $y_j\mathbf{x}$ . So every column lies in  $\text{span}(\mathbf{x})$ , and  $\text{col}(A) \subseteq \text{span}(\mathbf{x})$ . Since  $\mathbf{y} \neq \mathbf{0}$ , some  $y_j \neq 0$ , and then  $y_j\mathbf{x} \neq \mathbf{0}$  because  $\mathbf{x} \neq \mathbf{0}$  (Theorem 1.1.9). The list  $(\mathbf{x})$  is independent, since  $a\mathbf{x} = \mathbf{0}$  forces  $a = 0$  (Theorem 1.1.9), so  $\text{span}(\mathbf{x})$  has dimension 1, and  $\text{rank } A \leq 1$  by Theorem 1.6.7. On the other hand  $\text{rank } A \neq 0$ : a space of dimension 0 has the empty list as a basis, so it equals  $\text{span}() = \{\mathbf{0}\}$ , but  $\text{col}(A)$  contains the non-zero vector  $y_j\mathbf{x}$ . Hence  $\text{rank } A = 1$ .

( $\Rightarrow$ ) Suppose  $\text{rank } A = 1$ , and let  $(\mathbf{x})$  be a basis of  $\text{col}(A)$ ; then  $\mathbf{x} \neq \mathbf{0}$ . Each column  $\mathbf{a}_j \in \text{col}(A)$  equals  $y_j\mathbf{x}$  for some  $y_j \in F$ . Put  $\mathbf{y} = (y_1, \dots, y_n)$ . By the computation in ( $\Leftarrow$ ), the  $j$ -th column of  $\mathbf{x}\mathbf{y}^T$  is  $y_j\mathbf{x} = \mathbf{a}_j$ , so  $A = \mathbf{x}\mathbf{y}^T$ . If  $\mathbf{y} = \mathbf{0}$ , then  $A = \mathbf{0}$  would have rank 0, so  $\mathbf{y} \neq \mathbf{0}$ . This proves the claim.

**Exercise C3: The matrix  $i + j$ .**

For  $n \geq 1$ , let  $A_n \in M_n(\mathbb{R})$  have entries  $a_{ij} = i + j$ . Prove that  $\text{rank } A_1 = 1$  and  $\text{rank } A_n = 2$  for every  $n \geq 2$ . Hence find nullity  $A_n$ .

*Hint: write each row as a combination of two fixed row vectors.*

*Solution.* For  $n = 1$ ,  $A_1 = (2) \neq 0$ , so  $\text{rank } A_1 = 1$ . Let  $n \geq 2$ , and put  $\mathbf{1} = (1 \ 1 \ \dots \ 1)$  and  $\mathbf{d} = (1 \ 2 \ \dots \ n)$  in  $M_{1 \times n}(\mathbb{R})$ . Row  $i$  of  $A_n$  is  $(i+1 \ \dots \ i+n) = i\mathbf{1} + \mathbf{d}$ . So  $\text{row}(A_n) \subseteq \text{span}(\mathbf{1}, \mathbf{d})$ , and  $\dim \text{row}(A_n) \leq 2$  by Theorem 1.6.3.

Rows 1 and 2 are  $(2 \ 3 \ \dots)$  and  $(3 \ 4 \ \dots)$ . If  $c_1(\text{row } 1) + c_2(\text{row } 2) = \mathbf{0}$ , the first two entries give  $2c_1 + 3c_2 = 0$  and  $3c_1 + 4c_2 = 0$ . Subtracting the first from the second gives  $c_1 + c_2 = 0$ , so  $c_1 = -c_2$ , and then  $-2c_2 + 3c_2 = c_2 = 0$ , so  $c_1 = 0$ . Hence  $\text{row}(A_n)$  contains two independent vectors, and  $\dim \text{row}(A_n) \geq 2$  by Theorem 1.6.3.

So  $\dim \text{row}(A_n) = 2$ , and  $\text{rank } A_n = 2$  by Theorem 2.5.5. By Theorem 2.5.7, nullity  $A_n = n - 2$  for  $n \geq 2$ , and nullity  $A_1 = 0$ .

**Exercise C4: Counting right-hand sides over a finite field.**

Let  $F$  be a finite field with  $q$  elements, and let  $A \in M_{m \times n}(F)$  have rank  $r$ .

- Prove that exactly  $q^r$  vectors  $\mathbf{b} \in F^m$  make  $A\mathbf{x} = \mathbf{b}$  consistent.
- By counting the pairs  $(\mathbf{x}, \mathbf{b}) \in F^n \times F^m$  with  $A\mathbf{x} = \mathbf{b}$  in two ways, and using Theorem 2.1.3 but **not** Theorem 2.5.7, prove that  $\text{nul}(A)$  has exactly  $q^{n-r}$  elements. Deduce nullity  $A = n - r$ : a counting proof of Rank–Nullity over a finite field.

- Over  $\mathbb{F}_2$ , let  $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ . How many  $\mathbf{b} \in \mathbb{F}_2^3$  give a consistent system, and how many solutions does each such system have?

*Solution.*

- (a) By Theorem 2.1.1,  $A\mathbf{x} = \mathbf{b}$  is consistent exactly when  $\mathbf{b} \in \text{col}(A)$ . Let  $(\mathbf{c}_1, \dots, \mathbf{c}_r)$  be a basis of  $\text{col}(A)$ . By Theorem 1.5.1,  $(t_1, \dots, t_r) \mapsto t_1\mathbf{c}_1 + \dots + t_r\mathbf{c}_r$  is a bijection  $F^r \rightarrow \text{col}(A)$ : every vector of  $\text{col}(A)$  is hit by exactly one tuple. Since  $F^r$  has  $q^r$  elements, so does  $\text{col}(A)$ .
- (b) Let  $N = |\text{nul}(A)|$ . Each  $\mathbf{x} \in F^n$  determines exactly one  $\mathbf{b} = A\mathbf{x}$ , so there are  $q^n$  pairs. Now group the pairs by  $\mathbf{b}$ . If  $\mathbf{b} \notin \text{col}(A)$ , there are none, by Theorem 2.1.1. If  $\mathbf{b} \in \text{col}(A)$ , pick one solution  $\mathbf{p}$ ; by Theorem 2.1.3 the solutions are  $\mathbf{p} + \text{nul}(A)$ , and  $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$  is a bijection from  $\text{nul}(A)$  onto this set (it is onto by definition, and  $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$  gives  $\mathbf{h} = \mathbf{h}'$ ). So each of the  $q^r$  vectors  $\mathbf{b}$  of (a) accounts for exactly  $N$  pairs, and there are  $q^r N$  pairs in total. Hence  $q^r N = q^n$ , and  $N = q^{n-r}$ .  
Let  $d = \text{nullity } A$ . The argument of (a), applied to a basis of  $\text{nul}(A)$  of length  $d$ , gives  $N = q^d$ . So  $q^d = q^{n-r}$ . Since  $q \geq 2$ , distinct exponents give distinct powers of  $q$ , so  $d = n - r$ . This is [Theorem 2.5.7](#), proved here by counting alone.
- (c) Over  $\mathbb{F}_2$ ,  $R_3 \rightarrow R_3 + R_1$  gives row 3 equal to  $(0, 1, 1)$ , and then  $R_3 \rightarrow R_3 + R_2$  makes it zero. So an echelon form is  $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ , and  $\text{rank } A = 2$  by [Corollary 2.5.6](#). By (a), exactly  $2^2 = 4$  of the 8 vectors  $\mathbf{b}$  give a consistent system, and by [Theorem 2.5.8](#) (c) each has  $2^{3-2} = 2$  solutions. (The consistent  $\mathbf{b}$  are those with  $b_1 + b_2 + b_3 = 0$ . The rows of  $A$  add up to zero, so every consistent  $\mathbf{b}$  satisfies this equation; exactly 4 vectors of  $\mathbb{F}_2^3$  satisfy it, and 4 are consistent, so the two sets coincide.)